

Profile

AI Software Engineer at U.S. News & World Report specializing in building and scaling production-grade agentic AI systems and data platforms with Django, LangChain/LangGraph, real-time streaming APIs, and high-throughput vector search (OpenSearch/FAISS). Proven at turning ambiguous product requirements into reliable, compliant solutions — from semantic search to B2B analytics agents with live graph and CSV generation — while driving cross-functional delivery and operational excellence across product, data, and engineering domains.

Experience

- AI Software Engineer

Aug 2024 – Present

U.S. News & World Report, New York

- Architect/Maintain ByteSage [Django backend] driving AI products with LangGraph agent workflows
 - Implement RAG pipelines using OpenSearch/FAISS for semantic retrieval with metadata filtering
 - Deploy streaming endpoints; harden reliability across dev/sandbox/UAT/prod environments.
 - Implement industry grade guardrails and conditional routing for production-safe agentic tool execution.
- Software Engineer

May 2023 – May 2024

COAPS, Florida State University

- Built pipelines processing 1M+ marine data points into PostgreSQL (10+ TB)
 - Developed Django REST APIs serving 200k+ users/researchers with the marine data acquisition dashboards
 - Maintained the data extraction endpoints and supported access points.
- Graduate Research Assistant

Aug 2022 – Apr 2023

Florida State University

- Built and validated deep learning systems (PyTorch, TensorFlow) for scientific anomaly detection.
 - Reduced inference time 40% via custom CUDA kernels; boosted throughput 30% with GPU clustering
- Junior DevOps Engineer

Jan 2021 – Dec 2021

ACS Pvt. Ltd, Mumbai

- Maintained CI/CD pipelines (Jenkins, Docker, SonarQube); cut MTTR 80% with AWS CloudWatch
 - Automated infrastructure provisioning with Terraform, reducing deployment time by 60%
- Web Development Intern

Jun 2020 – Dec 2020

Cart Geek, Mumbai

- Implemented secure authentication flows and payment gateway integrations

Projects

- Cerebral Valley Agent Hackathon - Agent IDE Mind Map** (Nov 2025): Using Antigravity IDE's reasoning logs to generate mind map for tracking the development process with AI agents and creating an abstraction layer to rollback and branch out approaches in a UI dashboard
- Portfolio AI Chatbot** (2024–Present): RAG assistant with API calls, Chroma vector DB, LangChain for conversational resume queries and Tavily for web research reference for the user's query related to any comparison task for role based research based on my resume and career.
- Transformer from Scratch** (Feb–Apr 2024): Built text completion model implementing self-attention; tracked perplexity dynamics
- Medical Image Segmentation** (Aug–Dec 2023): Polyp detection model (PyTorch/OpenCV) achieving 0.92 DSC, 0.86 JC
- YouTube AI Channel**: Live coding sessions on LLM systems (Llama 3, Mistral) and production workflows

Skills

- Languages:** Python, C/C++ (CUDA), Java, JavaScript/TypeScript
- AI/ML:** LangChain, LangGraph, PyTorch, NumPy, TensorFlow, OpenCV, RAG, Vector Store Ingestion and Storage
- Data/Infra:** Django, PostgreSQL, Airflow, OpenSearch, FAISS, Docker, AWS, ChromDB
- Tools/Observability:** Git, Jenkins, REST APIs, CloudWatch, New Relic, Snyk

Education

- Florida State University — M.S. Computer Science

May 2024

GPA: 3.8/4.0
- NMIMS University — B.Tech Information Technology

Aug 2022

GPA: 3.3/4.0

Certifications & Achievements

- Python Certification from Udemy
- Recognized Hyperparameter determination experiment for transformer **[Attention is all you need]** based text completion model

Impact Snapshot

Project / Impact	Tech Stack	Company & Date
Production AI workflows with tool orchestration and industry grade guardrails	Django, LangChain, LangGraph, DRF APIs, NVIDIA NeMo Guardrails	U.S. News (Present)
Improved system observability and reduced mean time to detect and report outages.	New Relic, Slackbots, ByteSage [Django based AI Infrastructure]	U.S. News (Present)
Real-time analytics Marine Data serving 200k+ users/researchers across 125 schools	Python, PostgreSQL, Django REST Framework	COAPS (2023–2024)
Optimized model inference pipelines with custom CUDA kernels, cutting latency by 40% for interactive research uses.	PyTorch, TensorFlow, CUDA kernels	FSU (2022–2023)
Automated CI/CD smoketests reducing deployment risk 80%	Jenkins, Docker, AWS EC2/S3, CloudWatch	ACS (2021)